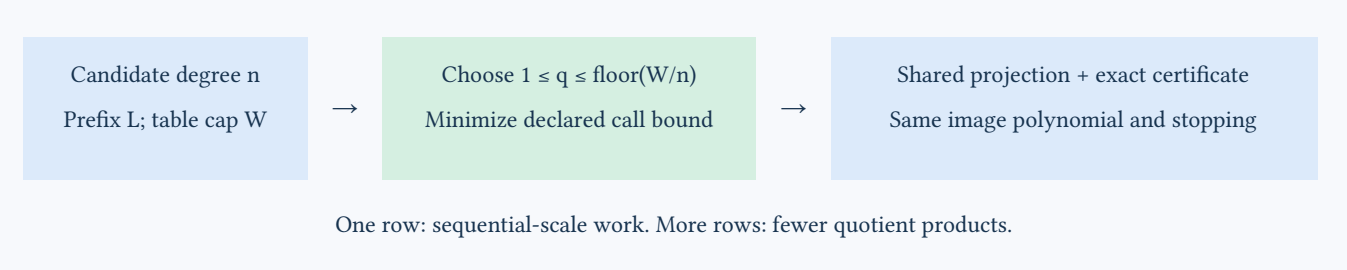
Workspace-adaptive polynomial projection

T-1822 · 2026-11-07 · derived algebra and unsigned producer checks

The adaptive recovery constructor now selects any legal positive polynomial block width under its short-table cap. A square-root-sized table is no longer required. The quotient-annihilation certificate and recovered image polynomial remain unchanged.



Declared objective. For prefix L, degree bound d=min(n,L), width q, let a=ceil(L/q) and b=ceil((d+1)/q). The short powers, projection and certificate require at most q+a+b-2 quotient products and exactly a projection convolutions. Select the legal q minimizing q+2a+b-2; ties prefer less table storage. This is a structural call bound, not a total-field-operation or timing optimum.

Derived continuum. Put K=floor(W/n), at least one. Across doubling prefixes stopping below4t, projection and certification cost O((sqrt(t)+t/K) M(n)+nt). Including setup and recurrence synthesis gives O(M(n)(log m+sqrt(t)+t/K)+nt+t^2+t log r). Table storage is bounded by W words and O(n sqrt(t)); other arrays and backend workspace are separate. Validation, output, parent checks and source evaluation remain additionally charged.

Native checks. 23,408 planner configurations satisfy admissibility, objective optimality and tie handling. 1,091 arithmetic configurations match direct power sums and full Horner remainders across workspace choices. Twenty-two below-one-row cases reject. A named F7 example formerly rejected by the square-root table cap now completes with one row and the same exact recurrence.

Recovery integration. The 6,522 small cases and three larger cases preserve complete image, envelope, moment and rejection-count results under both generous storage and exactly one row. All 63 public recoveries preserve scalar, route and query fields, with 511 source calls. The adapter uses a declared 64-coefficient-word short-table allowance on these fixtures. This does not measure total memory. Original 4,563-case cohort unchanged; no cloud work.

Evidence and next step. validation-01 compiled and passed without assertion failures; warnings retained. Receipts bind source dependencies, compiler and commands. Producer evidence has no independent signed verdict. See DERIVATION.md for assumptions and proof. Official FLINT modular-composition documentation supplies background; its generated-prose notice is recorded, and local identities are checked against exact references. KB unavailable. Next account for field arithmetic and source cost when choosing among recovery constructors.